GRC103 Advanced Cybersecurity Risk Quantification

Practical Lab 3B Cybersecurity Risk Quantification Lab

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Executive Summary

Problem

SecureMobile serves 500,000 customers and processes about 50,000 transactions a day, averaging $2,500 each (roughly $125M daily). A recent penetration test found three exploitable weaknesses in the platform: a database injection flaw, an API authentication bypass and a session hijacking flaw. If left untreated, they carry a combined expected loss of $35.0M a year. The largest, database injection, could expose all 500,000 customer profiles, and a fund-transfer attack could run for up to 48 hours before it is detected.

Recommendation

Approve $805,000 in Year 1 ($700,000 one-time plus $105,000 a year) for three security controls on the SecureMobile banking app. They are expected to cut annual expected losses from $35.0M to $8.0M, a 77% reduction.

Why now

Penetration testing found three serious weaknesses with a combined expected annual loss of $35.0M (probability × impact). One, a database injection flaw, accounts for $31.25M (89%) and could expose all 500,000 customer records.

What the money buys:

- **Web Application Firewall**: closes the database risk and cuts expected loss by $23.4M a year. Live within 30 days and it an annual cost of $175K.
- **Multi-factor authentication**: cuts account-takeover loss by $2.85M a year. It goes live within 75 days and has an annual cost of $230K
- **API Security Gateway** : lowest return, cutting $675K a year, but it closes the route to unauthorized fund transfers it goes live within 90 days. With an annual costs of $400K.

What remains:

There is an estimated $8.0M of expected loss, mostly database injection, stays after these controls. Over 12 months we will add code fixes, encryption and faster detection (today it takes 48 hours), and re-measure the risk each quarter.

Confidence:

These figures rest on worst-case loss assumptions and vendor-stated effectiveness. Even if the controls deliver half of what vendors claim, the program still returns about $15 for every $1 spent in Year 1.

Method and Key Assumptions

- **SLE** = quantity at risk × loss per unit
- **ALE** = SLE × ARO. ARO is the stated exploit probability
- **ALE share** = vulnerability ALE ÷ total ALE
- **Priority** is set by ALE thresholds: Critical ≥ $10M, High $1M–$10M, Medium $250K–$1M, Low < $250K
- **Risk reduction** = control effectiveness × ALE of the risk it targets
- **Year-1 cost** = initial investment + annual maintenance
- **ROI** = (risk reduction − Year-1 cost) ÷ Year-1 cost
- **Payback** = initial investment ÷ (risk reduction − annual maintenance)
- **Residual ALE** = ALE − risk reduction

Key assumptions

- Control effectiveness is taken from the stated figures and applied only to the vulnerability each control targets.
- SLEs are worst-case figures, assuming the whole asset is affected (all records, all sessions).
- ROI and payback measure expected value, not guaranteed outcomes.

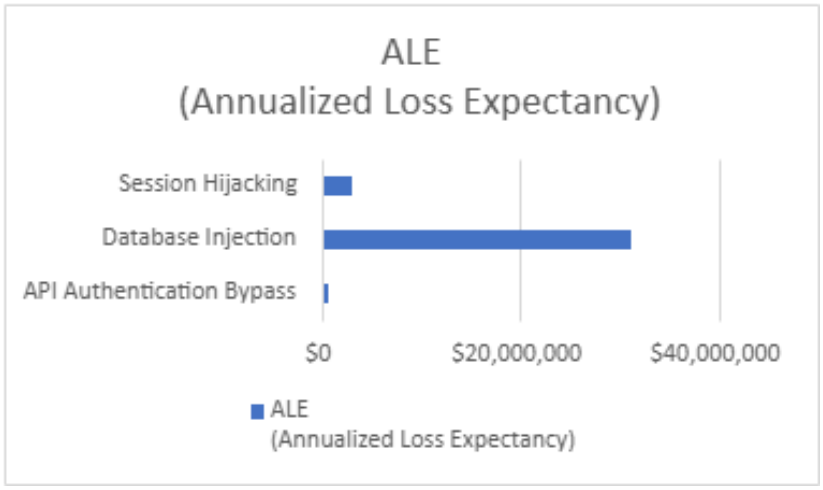
Required Visualizations:

Phase 1 : Risk Exposure

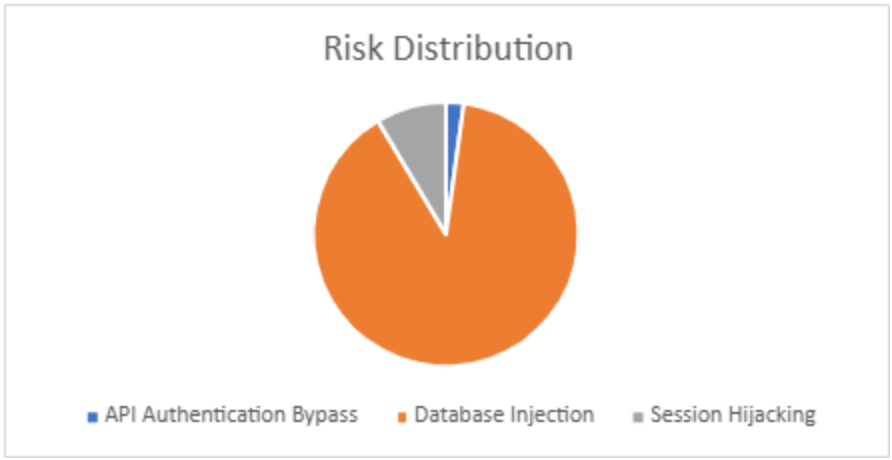
ALE Calculations and Rist Priotizations

Vulnerable Finding 1-3	Quantity at risk	Loss per unit (USD)	SLE (Single Loss Expectancy)	ARO (Annual Rate of Occurrence)	ALE (Annualized Loss Expectancy)	ALE total %	Priority
API Authentication Bypass	1	$5,000,000	$5,000,000	15%	$750,000	2.14%	Medium
Database Injection	500,000	$250	$125,000,000	25%	$31,250,000	89.29%	Critical
Session Hijacking	5,000	$1,500	$7,500,000	40%	$3,000,000	8.57%	High
Total					$35,000,000		

Bar chart comparing ALE for all vulnerabilities



Pie chart showing risk distribution



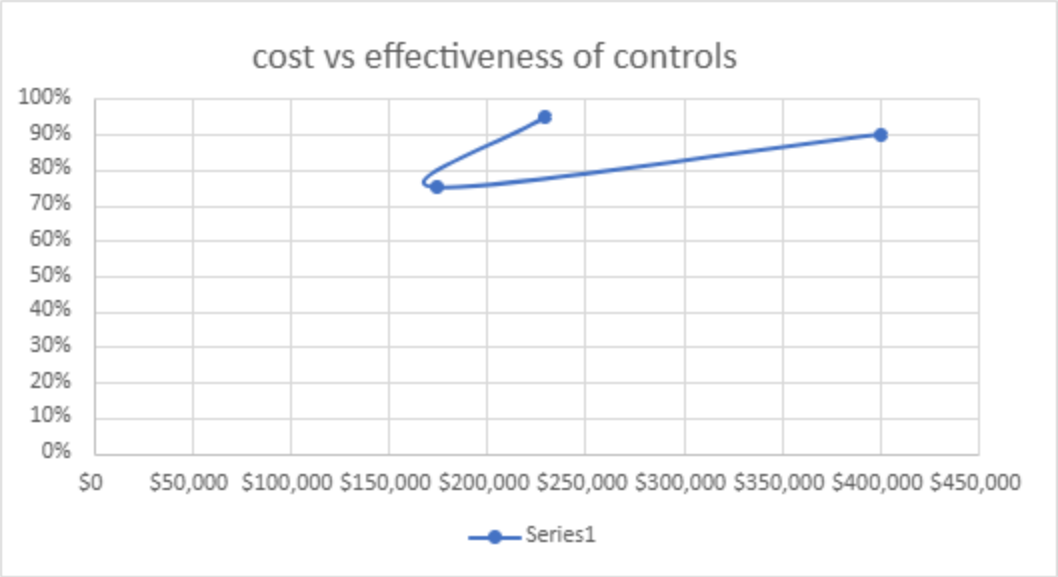
Phase 2: Control Evaluation

Cost-Benefit Analysis

Control	Targets	Initial investment	Annual maintenance	Effectiveness	Target ALE	Risk reduction per year	Year-1 cost	Year-1 ROI	Payback (years)	Payback (days)	3-yr total cost	3-yr ROI	Residual ALE of target
API Security Gateway	API Authentication Bypass	$350,000	$50,000	90%	$750,000	$675,000	$400,000	69%	0.5600	204.4	$500,000	305%	$75,000
Web Application Firewall	Database Injection	$150,000	$25,000	75%	$31,250,000	$23,437,500.00	$175,000	13293%	0.0064	2.3	$225,000	31150%	$7,812,500
MFA Enhancement	Session Hijacking	$200,000	$30,000	95%	$3,000,000	$2,850,000	$230,000	1139%	0.0709	25.9	$290,000	2848%	$150,000
Total		**$700,000**	**$105,000**		**$35,000,000**		$805,000				**$1,015,000**		**$8,037,500**

Note: Formulas available on the attached spreadsheet

Scatter plot showing cost vs. effectiveness of controls



Bar chart comparing ROI for all controls

Line graph showing risk reduction over time

Phase 3 Executive Reporting

Task 7: Risk Treatment

Horizon	Actions	Outcome
Immediate (0–30 days)	Emergency war-room and formal risk acceptance; virtual patch / input validation hotfix for Finding 2 deploy WAF; enhanced logging and alerting on API and database	Largest exposure cut by 75% by day 30

Short-term (90 days)	Roll out MFA; deploy API Security Gateway; session hardening (token rotation, session binding)	Finding 3 down 95%, Finding 1 down 90%
Long-term (1 year)	Penetration retest, database hardening (parameterized queries, encryption, tokenization) and reduce detection time from 48h to under 4h; quarterly risk re-quantification	Address $7.8M residual and validate vendor effectiveness claims

Conclusion

The assessment shows that SecureMobile carries $35.0M of expected annual loss, with database injection responsible for 89% of it ($31.25M). The three recommended controls (a Web Application Firewall, multi-factor authentication and an API Security Gateway) cost $805,000 in Year 1 and reduce expected loss by $26.96M (77%), leaving $8.04M of residual risk. All three controls give a positive return. The WAF is the strongest, so it should be deployed first, followed by MFA and then the API Gateway. Because the residual database risk is still significant, the controls should be paired with code-level fixes, encryption and faster detection, and the risk should be re-measured each quarter.

Attachments

Link to Spreadsheet